



OOP CLASS RELATIONSHIPS

Quiz

Association · Aggregation · Composition

12 Questions · Click through to reveal answers

Conceptual

Q1 · Q2 · Q3 — Test core understanding

Q1

CONCEPT

Question

What is the main difference between Aggregation and Composition? Give one real-life example of each.

→ Next slide for answer

Q1 — Answer

ANSWER

Aggregation: child can exist without parent.
Example: Teacher exists without a Department.

Composition: child cannot exist without parent.
Example: A Room cannot exist without a House.

Q2

CONCEPT

Question

In Association, does one class own the other? Explain with an example.

→ Next slide for answer

Q2 — Answer

ANSWER

No. Association is just a 'uses' relationship — no ownership.

Example: a Doctor uses a Stethoscope during a checkup, but neither owns the other. Both exist independently.

Q3

CONCEPT

Question

If you delete the parent object in Composition, what happens to the child?
What about in Aggregation?

→ Next slide for answer

Q3 — Answer

ANSWER

Composition → child is destroyed automatically with the parent.

Aggregation → child continues to exist. It was never owned by the parent.

Code Reading

Q4 · Q5 · Q6 — Identify from code

Q4

CODE

Question

What type of relationship is this – and how do you know?

```
class Wheel { ... };  
class Car {  
    Wheel w1, w2, w3, w4;  
};
```

→ Next slide for answer

Q4 — Answer

ANSWER

Composition.

The wheels are stored by value inside Car. They are created when Car is created and destroyed when Car is destroyed. Wheels have no independent existence.

Q5

CODE

Question

What type of relationship is this?

```
class Student { ... };  
class Library {  
    vector<Student*> members;  
};
```

→ Next slide for answer

Q5 — Answer

ANSWER

Aggregation.

Library stores pointers to Student objects. Students exist independently. Destroying the Library does NOT destroy the students.

Q6

CODE

Question

What type of relationship is shown here?

```
class Printer { public: void print() {...} };  
class Employee {  
    void doWork(Printer& p) { p.print(); }  
};
```

→ Next slide for answer

Q6 — Answer

ANSWER

Association.

Employee uses Printer only inside a function – it doesn't store it as a member variable. The link only exists while `doWork()` is running.

Scenarios

Q7 · Q8 · Q9 — Apply to real situations

Q7

SCENARIO

Question

A Bank Account has a Transaction History.
If the account is deleted, the transaction history should also be deleted.

What relationship is this?

→ Next slide for answer

Q7 — Answer

ANSWER

Composition.

Transaction History cannot meaningfully exist without the Bank Account it belongs to. They share a lifecycle — delete the account, delete its history.

Q8

SCENARIO

Question

A Playlist contains Songs.

If you delete a Playlist, the Songs should still exist in the music library.

What relationship is this?

→ Next slide for answer

Q8 — Answer

ANSWER

Aggregation.

Songs exist independently. A Playlist just groups references to them. Deleting the playlist doesn't mean the songs disappear.

Q9

SCENARIO

Question

A Customer places an Order at a restaurant.
The Customer and Order are separate systems and neither permanently belongs to the other.

What relationship is this?

→ Next slide for answer

Q9 — Answer

ANSWER

Association.

Customer and Order interact but neither owns the other. The Order system exists independently of the Customer object.

Tricky

Q10 · Q11 · Q12 — For advanced students

Q10

TRICKY

Question

True or False:
Aggregation and Composition are both types of Association.

Justify your answer.

→ Next slide for answer

Q10 — Answer

ANSWER

True.

All three describe relationships between classes. Association is the broadest category. Aggregation and Composition are more specific types of Association — the difference is only in ownership strength.

Q11

TRICKY

Question

A Human has a Brain.
A Human has a Friend.

Which relationship is which — and why?

→ Next slide for answer

Q11 — Answer

ANSWER

Human–Brain = Composition.

A brain cannot function without a human body. They share a lifecycle.

Human–Friend = Association.

Friends are independent people who use/interact with each other. Neither owns the other.

Q12

TRICKY

Question

In C++, what is the single biggest clue in code that tells you whether a relationship is Composition vs Aggregation?

→ Next slide for answer

Q12 — Answer

ANSWER

How the child is stored:

ChildClass obj; → Composition (by value, owned)

ChildClass* obj; → Aggregation (by pointer, not owned)

The pointer means 'I know about it but don't own it.'



End of Quiz

Association · Aggregation · Composition